

ECON 311: Intermediate Macroeconomic Theory

Lecture 1: Introduction

Introduction to Macroeconomic Analysis

Measuring Aggregate Economic Activity

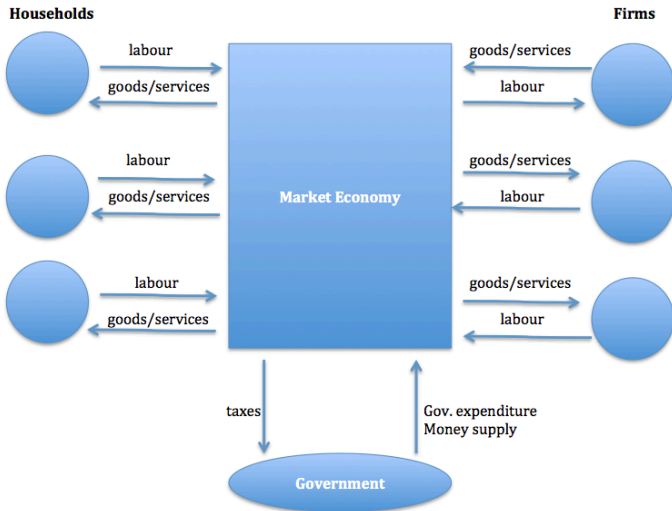
Long Run

Short Run

Definitions

- **Economics** is the study of how to manage scarce resources.
 - ▶ Economic agents decide how to manage their scarce resources:
 - ▶ E.g. workers choose how to allocate their time between labour and leisure.
 - ▶ E.g. consumers choose how to allocate their income across different goods and services.
 - ▶ E.g. firms make production decisions.
- **Macroeconomics** studies the interaction of the decisions of economic agents at the national level.

Definitions



Macroeconomic Approach

1. Document a set of empirical regularities or stylized facts about economies.
2. Develop a model that focus on relevant economic relations.
3. Use data to determine the model's parameters and exogenous variables
4. Measure how well the model explain endogenous variables given parameters and exogenous variables.

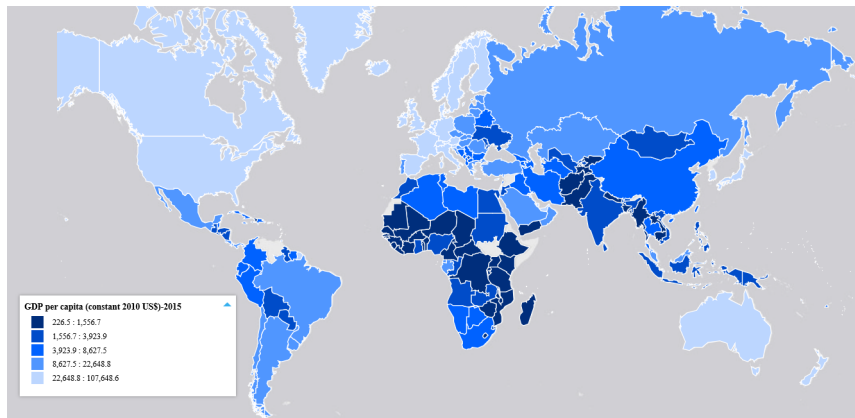
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GDP across countries



Source: World Development Indicators

GDP across countries

Canada

GDP growth	1.7%
GDP per head	\$57,190 (PPP: \$67,090)
Inflation	2.2%
Budget balance (% GDP)	-1.0
Population	40.1m

Mexico

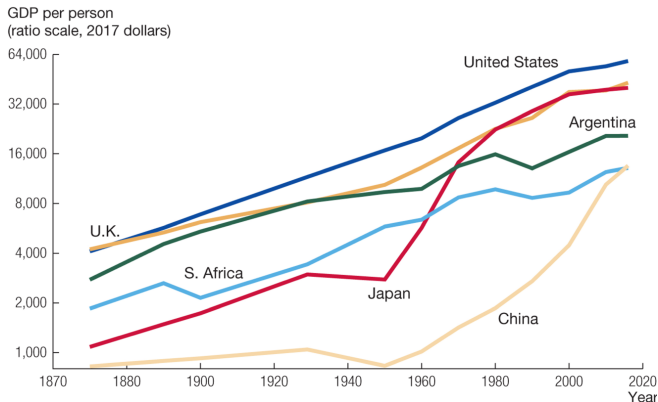
GDP growth	1.4%
GDP per head	\$14,290 (PPP: \$27,060)
Inflation	4.0%
Budget balance (% GDP)	-4.0
Population	131.9m

United States

GDP growth	1.7%
GDP per head	\$86,190 (PPP: \$86,190)
Inflation	3.0%
Budget balance (% GDP)	-6.9
Population	347.3m

Source: The Economist

GDP across time and countries



Source: The Maddison-Project, www.ggd.net/maddison/.
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Source: Macroeconomics, Jones 4E

Measuring Aggregate Economic Activity

In **lecture 2**, we will review how economists measure aggregate economy activity.

- In particular, we will review ...
 1. different approaches to measure the value of output (ECON 101-Review)
 2. how to compare economic activity across time (price indexes):
 - ▶ Chain-Weighting Approach to measure changes in real GDP over the long-run
 3. how to compare economic activity across countries:
 - ▶ using PPP-exchange rates

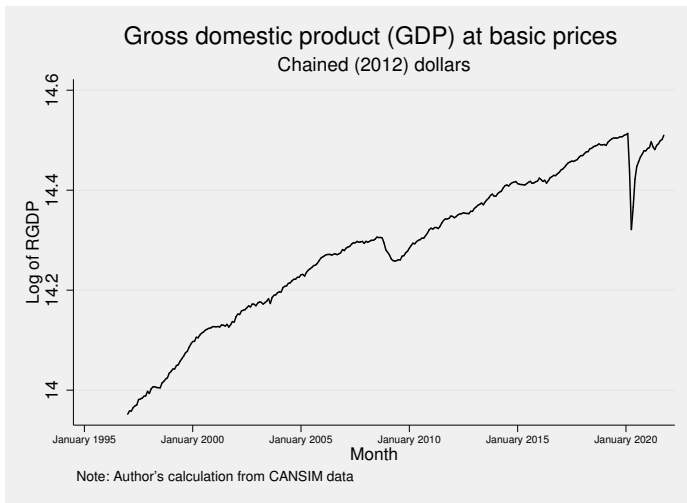
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Real GDP per Capita



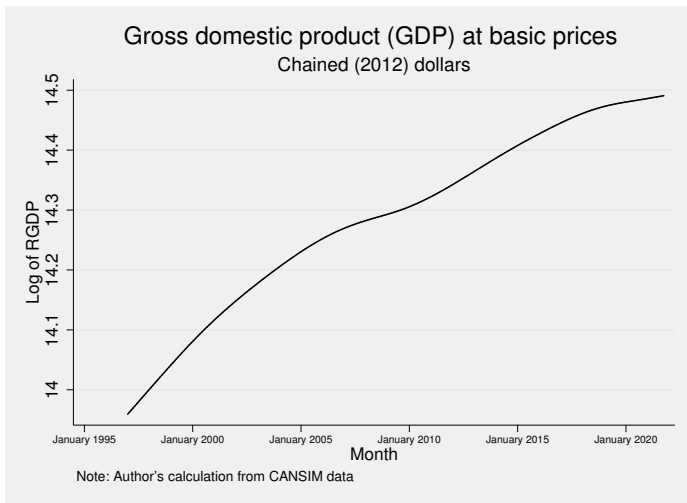
Decomposing Real GDP per Capita

- Annual average growth of real GDP per capita 1960–2017 $\approx 2.01\%$
- The series is composed of a trend (t) and cyclical (c) components:

$$y_t = c_t + t_t$$

- ▶ **Trend component:** Underlying structure of the series
- ▶ **Cyclical component:** Temporary shocks

Real GDP per Capita – Trend



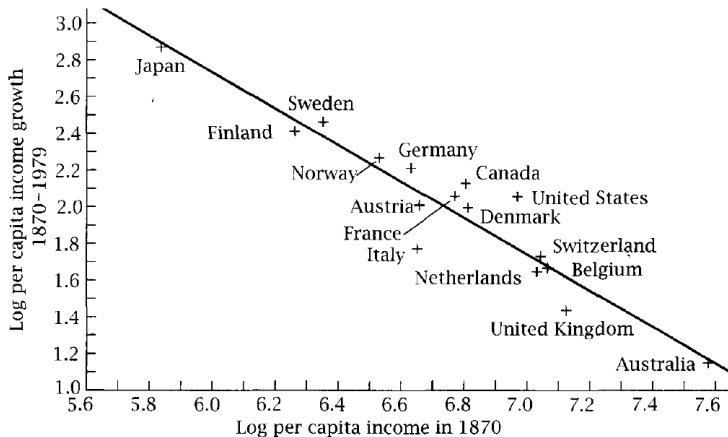
Real GDP per Capita Across Time and Countries

Country	Period	Real GDP per Person at Beginning of Period*	Real GDP per Person at End of Period*	Growth Rate (per year)
Japan	1890–2010	\$1 563	\$35 861	2.65%
Brazil	1900–2010	809	11 311	2.43
Mexico	1900–2010	1 204	14 783	2.31
China	1900–2010	745	7 747	2.15
Germany	1870–2010	2 271	39 569	2.06
Canada	1870–2010	2 469	39 528	2.00
United States	1870–2010	4 166	48 635	1.77
Argentina	1900–2010	2 384	15 937	1.74
India	1900–2010	702	3 431	1.45
United Kingdom	1870–2010	4 999	36 695	1.43
Indonesia	1900–2010	926	4 306	1.41
Pakistan	1900–2010	766	2 843	1.20
Bangladesh	1900–2010	648	1 854	0.96

*Real GDP is measured in 2010 Canadian dollars.

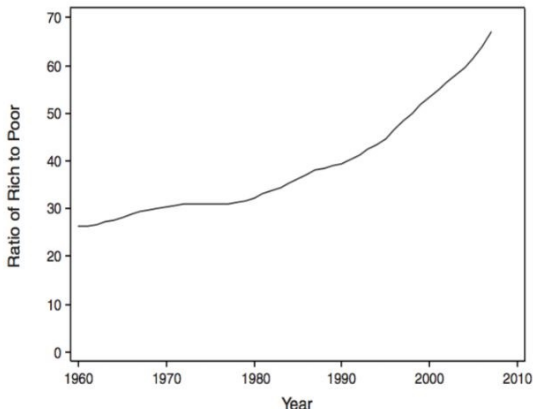
Source: Mankiw, N., Kneebone, R., & McKenzie, K. (2017). *Principles of Macroeconomics* (7th Canadian ed.). Toronto, ON: Nelson Education

Real GDP per Capita Across Time and Countries



Real GDP per Capita Across Time and Countries

Figure 1 GDP per Capita Ratio of Rich to Poor



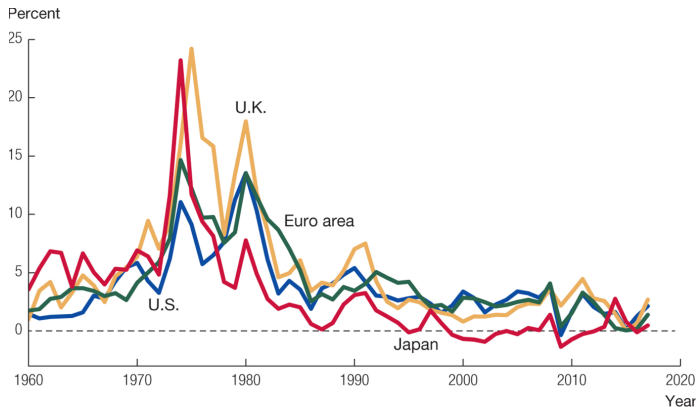
Notes: The ratio refers to the average of the richest 5 percent of countries to the poorest 5 percent of countries in each year. Since the sample contains 101 countries, these are averages of 5 countries. Source: Restuccia 2011

Long-Run Economic Growth

Lucas (1988): “Once one starts to think about economic growth, it is hard to think about anything else”

- In **lectures 3-10**, we study the determinants of long-run growth:
 1. Basic facts about long-run growth across countries
 2. Disaggregate cross-country differences in output per-capita into differences in resources to produce (growth accounting)
 3. Study two important long-run growth model with very different implications for long-run growth:
 - ▶ Solow Growth Model
 - ▶ Romer Growth Model

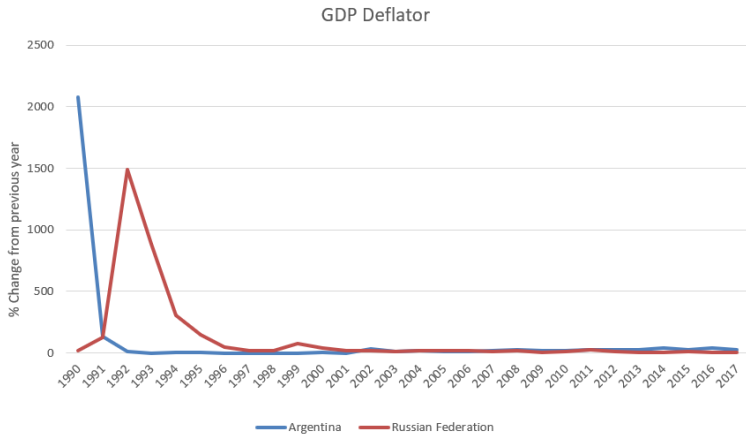
Money and Inflation



Source: FRED. The inflation rate measures the percentage change in consumer prices.
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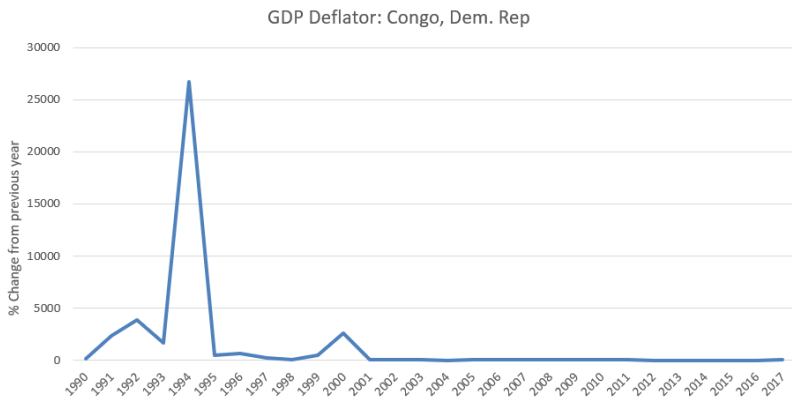
Source: Macroeconomics, Jones 4E

Money and Inflation



Source: World Development Indicators

Money and Inflation



Source: World Development Indicators

Money and Inflation

- In **lecture 13**, we briefly discuss the topic of inflation in the long-run:
 1. Determinants of long-run inflation
 2. The Quantity Theory of Money (relationship between the growth of money and inflation)

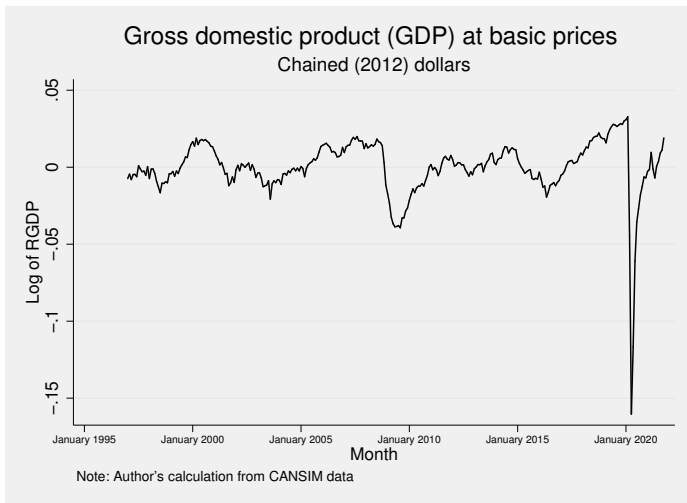
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Real GDP per Capita – Cyclical Component

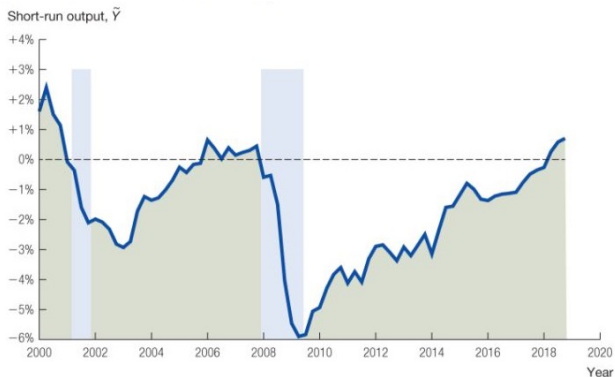


Short-Run Economic Fluctuations

- In **lectures 13-21**, we study the determinants of short-run economic fluctuations:
 1. Basic facts about short-run deviations from potential output
 2. Shocks to the economy, inflation and government short-run economic policy
 3. The central macroeconomics model of short-run economic fluctuations (AS-AD model)

Recent Economic Crisis

- In our last two lectures (22 and 23), we turn our attention to studying different dimensions of the two most recent economic crisis
 - ▶ The Great Recession of 2008



Source: Macroeconomics, Jones 4E

Recent Economic Crisis

- In our last two lectures (22 and 23), we turn our attention to studying different dimensions of the two most recent economic crisis

► The COVID-19 Recession

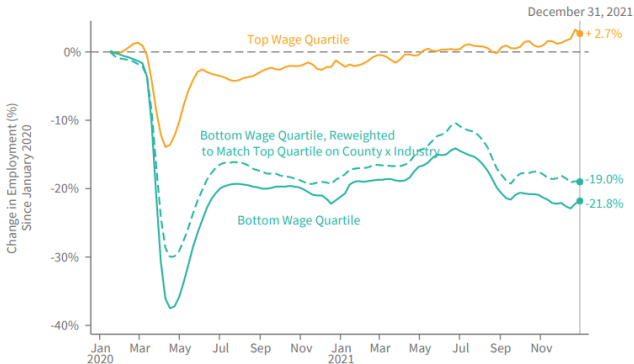


FIGURE: Changes in Employment by Wage Quartile, Source: Opportunity Insight